Assignment_02_CO_01

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QUESTIONS

- Q1 Print a hollow diamond pattern — 5 marks
- Q2 Design a class to calculate parking charges — 5 marks
- Q3 Print a number pattern with consecutive values — 5 marks
- Q4 Check whether a number is a spy number — 5 marks
- Q5 Print a butterfly pattern — 5 marks

5/5 answered

Q1 Print a hollow diamond pattern — 5 marks

Use nested loops to print a diamond of five rows in the upper half and four rows in the lower half.

```
*
* *
* *
* *
* *
* *
* *
*
```

Your Code

```cpp
#include <iostream>
using namespace std;
int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= (n - i + 1); j++) {
            cout << "*";
        }
        for (int j = 1; j <= 2 * (i - 1); j++) {
            cout << " ";
        }
        for (int j = 1; j <= (n - i + 1); j++) {
            cout << "*";
        }
        cout << endl;
    }

    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= (n - i + 1); j++) {
            cout << "*";
        }
        for (int j = 1; j <= 2 * (i - 1); j++) {
            cout << " ";
        }
        for (int j = 1; j <= (n - i + 1); j++) {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

Input

stdin input

Output

```
**********
****  ****
***    ***
**      **
*        *
**      **
```

Submitted

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SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

JANA LAHARI
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5/5 answered

Q2 Design a class to calculate parking charges

5 marks

Create a Parking class. Charge ₹30 for the first hour and ₹20 for every additional hour. Any fraction of an hour counts as a full hour.
TC_1: 2 hours and 15 minutes →Parking charge: Rs.70. TC_2: 0 hours and 30 minutes →Parking charge: Rs.30 .TC_3: 2 hours and 60 minutes → Invalid duration

Your Code

```cpp
#include <iostream>
using namespace std;
class Parking {
public:
    int h, m;
    void calculate() {
        int t = h * 60 + m;
        if (h < 0 || m < 0 || m >= 60 || t == 0)
            cout << "Invalid duration";
        else
            cout << "Parking charge: Rs."
                 << 30 + (t > 60 ? (t - 1) / 60 * 20 : 0);
    }
};
int main() {
    Parking p;
    cin >> p.h >> p.m;
    p.calculate();
    return 0;
}
```

Input

```
2 15
```

Output

```
Parking charge: Rs.70
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

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- Q5 Print a butterfly pattern 5 marks

5/5 answered

Q3 Print a number pattern with consecutive values — 5 marks

Print the following pattern for four rows.

```
1
2 3
4 5 6
7 8 9 10
```

Your Code

```cpp
#include <iostream>
using namespace std;
int main() {
    int n = 1;
    for (int i = 1; i <= 4; i++) {
        for (int j = 1; j <= i; j++)
            cout << n++ << " ";
        cout << endl;
    }
    return 0;
}
```

Input

```
stdin input
```

Output

```
1
2 3
4 5 6
7 8 9 10
```

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QUESTIONS

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- Q5 Print a butterfly pattern 5 marks

5/5 answered

Q4 Check whether a number is a spy number — 5 marks

A number is a spy number if the sum of its digits equals the product of its digits.
TC_1:1124 → Spy number. TC_2:124 → Not a spy number. TC_3:-123 → Invalid input

Your Code

```cpp
#include <iostream>
using namespace std;
int main() {
    int n, sum = 0, prod = 1, d;
    cin >> n;
    if (n < 0) {
        cout << "Invalid input";
        return 0;
    }
    int t = n;
    while (t > 0) {
        d = t % 10;
        sum += d;
        prod *= d;
        t /= 10;
    }

    if (sum == prod)
        cout << "Spy number";
    else
        cout << "Not a spy number";

    return 0;
}
```

Input

```
1124
```

Output

```
Spy number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

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5/5 answered

Q5 Print a butterfly pattern — 5 marks

Print a butterfly pattern using nested loops.

```
* *
** **
*** ***
**** ****
**********
**** ****
*** ***
** **
* *
```

Your Code

```cpp
#include <iostream>
using namespace std;
int main() {
    for(int i=1;i<=5;i++) {
        for(int j=1;j<=i;j++) cout<<"*";
        if(i<5) cout<<" ";
        for(int j=1;j<=i;j++) cout<<"*";
        cout<<endl;
    }
    for(int i=4;i>=1;i--) {
        for(int j=1;j<=i;j++) cout<<"*";
        cout<<" ";
        for(int j=1;j<=i;j++) cout<<"*";
        cout<<endl;
    }
    return 0;
}
```

Input

```
stdin input
```

Output

```
* *
** **
*** ***
**** ****
**********
**** ****
```

Submitted

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